

The Conductor Census: Two Schemes, One Genus

Exact character histograms across the four rungs $N = 7/9/15/30$

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The “Dynamic Principle” program — the hodge-laboratory repository

Standalone theorem monograph · Monograph 20/20

1 Setting

While monograph 1/20 derived the genus of the Fermat curve topologically (Riemann–Hurwitz), here the genus is obtained *arithmetically* — as the total of the conductor census of the eigencharacters. The census is the arithmetic skeleton of the ladder: it determines the multiplicities h_d with which the conductors $d \mid N$ enter every Coleman–Selberg product of the program (certificates G and H), and it works identically on all four rungs $N = 7, 9, 15, 30$. Protocol V1 executes *two independent census schemes* — the direct one and the Möbius one — and demands their bit-for-bit agreement; this monograph proves that both schemes count the same thing and collects the exact histograms of every rung.

Definition 1 (eigencharacter and conductor). An eigencharacter of level N is a pair (a, b) of integers with $1 \leq a, 1 \leq b, a + b \leq N - 1$; it defines a holomorphic differential $\omega_{a,b} = x^a y^b dx/y^b$ on the Fermat curve $X_N: x^N + y^N = 1$. The conductor of a character is $d = N / \gcd(N, a, b)$ — the smallest level through which the character factors. The census is the histogram h_d of the conductors; $c(k) = (k - 1)(k - 2)/2$ is the total number of eigencharacters of level k .

Theorem 2 (the triangle). $c(N) = (N - 1)(N - 2)/2$, and the census total equals the genus: $\sum_{d \mid N} h_d = c(N) = (N - 1)(N - 2)/2 = g$.

Proof. The number of pairs $1 \leq a, b$ with $a + b \leq N - 1$ is the count of lattice points of the integer triangle: for $a = 1, \dots, N - 2$ there are exactly $N - 1 - a$ values of b , giving $\sum_{a=1}^{N-2} (N - 1 - a) = 1 + 2 + \dots + (N - 2) = (N - 1)(N - 2)/2$. Each pair yields a basis holomorphic differential (classical for the Fermat curve, monograph 1/20), so $c(N) = g$ — the arithmetic side of the genus formula, independent of Riemann–Hurwitz. Since every character has exactly one conductor (lemma 3 below shows the conductors divide N), the census total is $c(N)$. $\square$

Theorem 3 (the lifting lemma and Möbius inversion). For every $d \mid N$ the map $(a', b') \mapsto (\frac{N}{d}a', \frac{N}{d}b')$ is a bijection between the eigencharacters of level d and the eigencharacters of level N whose conductor divides d . Consequently,

$$\sum_{d' \mid d} h_{d'} = c(d), \quad \text{and by Möbius inversion} \quad h_d = \sum_{m \mid d} \mu(m) c(d/m).$$

In particular h_d does not depend on the ambient level N (only $d \mid N$ matters): $h_d(N) = h_d(d)$ — the self-similarity of the ladder: every conductor carries a complete smaller stand.

Proof. Well-definedness. Let (a', b') be a character of level d . Put $a = \frac{N}{d}a', b = \frac{N}{d}b'$: then $1 \leq a, b$ and $a + b = \frac{N}{d}(a' + b') \leq \frac{N}{d}(d - 1) = N - \frac{N}{d} \leq N - 1$, so (a, b) is a character of level N . Its conductor: $\gcd(N, a, b) = \frac{N}{d} \gcd(d, a', b')$, hence $d(a, b) = N / \gcd(N, a, b) = d / \gcd(d, a', b') = d(a', b') \mid d$.

Surjectivity with a section. If a character (a, b) of level N has conductor $d' \mid d$, then $\gcd(N, a, b) = \frac{N}{d'}$ is a multiple of $\frac{N}{d}$, whence $\frac{N}{d} \mid a, b$ and the pair $a' = \frac{d}{N}a, b' = \frac{d}{N}b$ is integral; the bounds are preserved by the reverse of the well-definedness argument, and (a', b') is a character of level d with conductor d' that restores (a, b) . Injectivity is obvious. Therefore $\sum_{d' \mid d} h_{d'}(N) = c(d)$ — the total number of characters of level d .

Inversion. The divisor lattice of d is a finite distributive lattice, and its Möbius function is the classical μ ; Möbius inversion on the divisor lattice yields the second formula. The right-hand side depends only on d , so h_d is the same at every level N divisible by d : at the level d the formula produces $h_d(d)$. The self-similarity is proved. $\square$

Theorem 4 (the censuses of the four rungs). *The exact conductor histograms:*

$$\begin{aligned} N = 7 : & \quad \{7: 15\}, & 15 = 15; \\ N = 9 : & \quad \{3: 1, 9: 27\}, & 28 = 1 + 27; \\ N = 15 : & \quad \{3: 1, 5: 6, 15: 84\}, & 91 = 1 + 6 + 84; \\ N = 30 : & \quad \{3: 1, 5: 6, 6: 9, 10: 30, 15: 84, 30: 276\}, & 406 = 1 + 6 + 9 + 30 + 84 + 276. \end{aligned}$$

Each row is verified by both schemes, including the explicit values $h_6 = c(6) - c(3) - c(2) + c(1) = 10 - 1 - 0 + 0 = 9$, $h_{10} = c(10) - c(5) - c(2) + c(1) = 36 - 6 = 30$ and $h_{30} = c(30) - c(15) - c(10) - c(6) + c(5) + c(3) + c(2) - c(1) = 406 - 91 - 36 - 10 + 6 + 1 = 276$.

Proof. The direct scheme. $N = 7$ is prime: $\gcd(7, a, b) = 1$ for every character of the triangle — the single conductor 7, $h_7 = c(7) = 15$. $N = 9$: conductor 3 means $\gcd(9, a, b) = 3$, i.e. $a, b \in \{3, 6\}$, and $a + b \leq 8$ leaves $(3, 3)$: $h_3 = 1$; the rest $h_9 = 28 - 1 = 27$. $N = 15$: conductor 3 — $\gcd(15, a, b) = 5$, i.e. $a, b \in \{5, 10\}$ with $a + b \leq 14$ — only $(5, 5)$: $h_5 = 1$; conductor 5 — $\gcd = 3$, i.e. $a, b \in \{3, 6, 9, 12\}$ with $a + b \leq 14$ — the six pairs $(3, 3), (3, 6), (6, 3), (3, 9), (9, 3), (6, 6)$: $h_5 = 6$; the rest $h_{15} = 91 - 1 - 6 = 84$. $N = 30$ — conductor 6: $\gcd(30, a, b) = 5$, so a, b are multiples of 5 with $a + b \leq 29$ and $\gcd$ exactly 5 (not 10, not 15) — nine pairs; the Möbius scheme gives the same $h_6 = 9$; likewise $h_{10} = 30$, $h_{15} = 84$ (self-similarity: $h_{15}(30) = h_{15}(15)$) and $h_{30} = 276$.

The Möbius scheme. The formula of theorem 3 with the explicit values c : $c(2) = 0$, $c(3) = 1$, $c(5) = 6$, $c(6) = 10$, $c(9) = 28$, $c(10) = 36$, $c(15) = 91$, $c(30) = 406$; the computations are displayed in the statement — every conductor matches the direct scheme pairwise. $\square$

Theorem 5 (two schemes, one total). *The direct scheme (enumerating the pairs with $d = N / \gcd(N, a, b)$) and the Möbius scheme (inclusion–exclusion over the divisors) produce bit-for-bit identical histograms at every level $N = 7, 9, 15, 30$ — an integer identity, a deterministic check without thresholds.*

Proof. By theorem 3 both schemes compute the same function h_d : the direct one by definition (every pair lands in exactly one conductor class), the Möbius one as the unique solution of the convolution equation $\sum_{d'|d} h_{d'} = c(d)$ on the divisor lattice. The formal coincidence leaves no room for discrepancies; the machine execution (protocol V1) is a fingerprint against implementation errors: two independent code paths must produce identical dictionaries $\{d: h_d\}$, which is exactly what the bit-wise comparison asserts. $\square$

Remark 6. The histograms compress quadratically with the arithmetic of the level: the prime 7 has one conductor; $9 = 3^2$ has two (the “hermit” $(3, 3)$ appears — the trace of level 3); $15 = 3 \cdot 5$ has three; $30 = 2 \cdot 3 \cdot 5$ has six — every divisor of the level except 1 and 2 (the triangle of level d is empty only for $d \leq 2$). The self-similarity $h_d(N) = h_d(d)$ turns the ladder into a fractal family of stands: a larger rung contains all smaller ones as conductor layers, and every summand of $406 = 1 + 6 + 9 + 30 + 84 + 276$ is itself the census of a smaller stand (its proper conductors subtracted). It is the multiplicities h_d that enter the exponents of the Coleman–Selberg product of certificate H — the census closes the Γ -architecture of the program.

Protocol data: The census — a summary of the four rungs

$N = 7$: genus 15, $\{7: 15\}$, a single conductor (prime level). $N = 9$: genus 28, $\{3: 1, 9: 27\}$, the diagonal character $(3, 3)$. $N = 15$: genus 91, $\{3: 1, 5: 6, 15: 84\}$. $N = 30$: genus 406, $\{3: 1, 5: 6, 6: 9, 10: 30, 15: 84, 30: 276\}$. All rows confirmed by two schemes (V1); the totals

equal $(N-1)(N-2)/2$ exactly; the self-similarity $h_d(N) = h_d(d)$ (e.g. $h_{15}(30) = 84 = h_{15}(15)$).
Reference: `baseline_v1_v9.json`, block `V1_census`.

Summary of results: summary

The arithmetic of the genus is placed on its own foundation: the character triangle, the lifting lemma with the conductor bijection, Möbius inversion, the self-similarity $h_d(N) = h_d(d)$, the exact histograms of all four rungs and the bit-for-bit agreement of the two schemes. The genus is *counted*, not postulated; the multiplicities h_d are the input of the Γ -normalization of certificates G/H .

Verification: how to reproduce

`python3 laboratory.py` → protocol V1 (the census, two schemes, levels 7/9/15/30);
`--check-baseline` cross-checks the histograms and the genera against the reference; Lean: the `genus_fermat` block; batch mode: the `census` item of a scenario.